6 CARRIER TRANSPORT

Net flow of electrons and holes in a semiconductor generate currents, and the process by which these charged particles move is called transport.

The carrier transport phenomena is the foundation for determining the current-voltage characteristic of semiconductor devices.

- There are two basic transport mechanisms in a semiconductor crystal
  - Drift : movement of charge due to electric fields
  - Diffusion : flow of charge due to concentration gradients
- In addition, the electrons and holes are in constant motion due to thermal energy. However, the motion due to the thermal energy is random and does not lead to current flow.

6.1 Thermal motion

Above 0 K, there is always random motion of particles due to thermal energy.

- The higher the temperature, the greater the thermal energy causing the thermal motion
- At temperature T, the average thermal energy of particles in 3 dimensions (3-D) is

$$E_{thermal\,(3-D)} = \frac{3}{2} k_B T \qquad \textbf{(6.1)}$$

(1/2 $k_B T$ per degree of freedom)

- The average thermal velocity v_{th} of the particle can be obtain by equating eqn. (6.1) to the kinetic energy of the motion

$$\frac{3}{2}k_B T = \frac{1}{2}m{v_{th}}^2 \qquad \therefore v_{th} = \sqrt{\frac{3k_B T}{m}} \quad \textbf{(6.2)}$$

- From eqn. (6.2), the average thermal velocity of electrons at room temperature (T = 300 K) is about 10^5 m/s (100 km/s). Here, we have used the electron rest mass.
    - this electron kinetic energy corresponds only to several hundredths of an electron volt, and it is not sufficient for electron even to leave semiconductor crystal
- In the absence of any external forces/influences, the electron (or hole) takes part only in the random thermal motion
- there is no directed motion whatsoever, there being no fixed direction which the electron (or hole) would prefer.

- although electrons (or holes) move randomly with a very great velocity, the velocity of the directed motion equals zero.

Fig. 6.1 illustrates the motion of a hole in a semiconductor with and without an electric field ξ.

  - without the electric field, there is not net movement of the hole
  - with the electric field, there is net movement of the hole

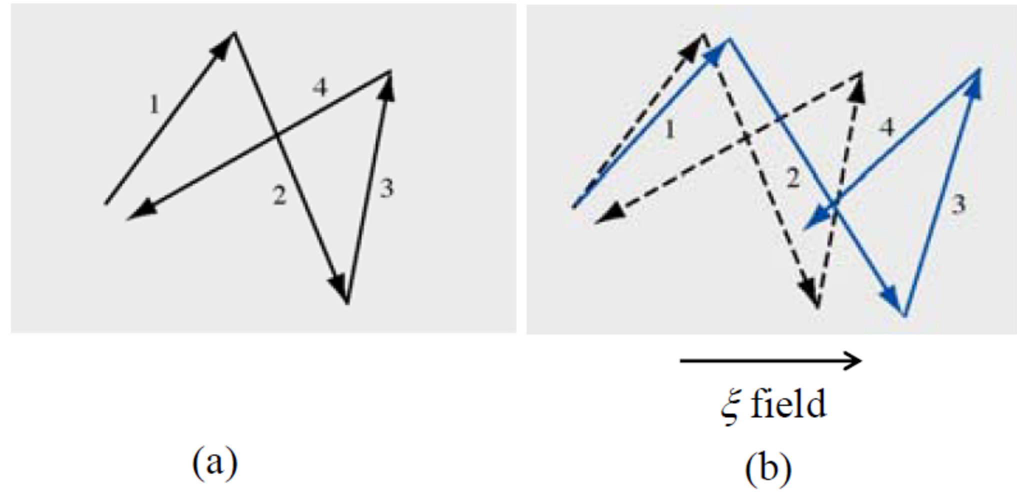


Fig. 6.1 Typical motion behavior of a hole in a semiconductor a) without an electric field and b) with an electric field

6.2 Drift current

An electric field applied to a semiconductor will produce a force on electrons and holes so that they will experience net acceleration and net movement.

- Directed motion is added on top of the thermal random motion of the carriers. This is illustrated in Fig. 6.1.b.

- Net movement of charge due to an electric field is called drift

- Net drift of charges give rise to a drift current

6.2.1 Mobility

Let us first consider the motion of a hole in the presence of an electric field ξ. The equation of motion is

$$F = q\xi = m_p^* a \qquad \textbf{(6.3)}$$

Here, F is the force experienced by the hole, q is the magnitude of the electric charge, a is the acceleration, ξ is the electric field and m_p^* is the effective mass of the hole

- If the electric field ξ is constant with time (i.e. does not change with time), velocity will increase linearly with time. Remember, the velocity at time t is

$$v(t) = v_0 + at \qquad \textbf{(6.4)}$$

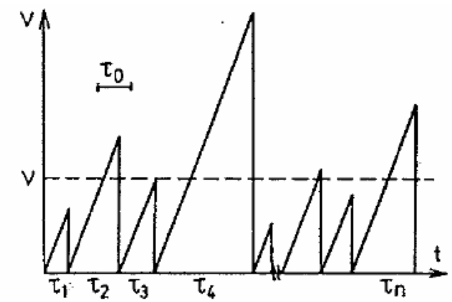
where v_0 is the initial velocity (velocity at $t = 0$)

- however, charged carriers in a semiconductor will experience collisions with
  - thermally vibrating lattice atoms
  - ionized impurity atoms and
  - defects
- after a collision the carrier can move in any direction, and this means that the velocity of directed motion after the collision is equal to zero. It follows, from eqn. (6.4), the velocity in the directed motion at time t is

$$v(t) = at \qquad \textbf{(6.5)}$$

- Let us denote the time between collisions be τ. Since the collisions are quite accidental, the times between collisions τ can also be quite different.

  The velocity of the directed motion of a charge carrier $v(t)$ in the presence of an electric field ξ versus time t is shown in Fig. 6.2.



The velocity of the directed motion of the electron (or the hole) in the electric field versus time at random scatterings.

Fig. 6.2

- Although the times between collisions are not the same (as shown in Fig. 6.2), there is an average time between collisions which may be denoted by τ_c. Consequently, there is corresponding average velocity in the directed motion, called drift velocity, v_d. From eqn. (6.5), we then have

$$v_d = a\,\tau_c \qquad \textbf{(6.6)}$$

- For a hole, we may then use the expression in eqn. (6.3) to substitute the acceleration a in eqn. (6.6). We obtain the drift velocity for hole

$$\boxed{v_{dp} = \frac{q\,\tau_{cp}}{m_p^*}\xi} \qquad \textbf{(6.7)}$$

where τ_{cp} is the mean time between collisions for holes

  - Note that in deriving eqn. (6.7) we have assumed low electric fields where τ_{cp} does not depend on ξ.

In eqn. (6.7), the hole drift velocity v_{dp} is directly proportional to the electric field ξ. We can define the hole mobility μ_p by writing

$$\boxed{v_{dp} = \mu_p \, \xi} \quad \textbf{(6.8)}$$

where

$$\boxed{\mu_p = \frac{q \, \tau_{cp}}{m_p^{*}}} \quad \textbf{(6.9)}$$

- mobility describes how well a particle will move due to an electric field
- unit of mobility is usually expressed in $cm^2/V\text{-}s$

We can apply the same analysis for electrons. Since electrons are negatively charged, the equation of motion of an electrons in the presence of an electric field ξ is

$$F = -q\xi = m_n^* a \qquad \textbf{(6.10)}$$

- The electron drift velocity is

$$v_{dn} = a\,\tau_{cn} \qquad \textbf{(6.11)}$$

where τ_{cn} is the mean time between collisions for electrons.

From eqn. (6.10), we obtain

$$\boxed{v_{dn} = -\frac{q\,\tau_{cn}}{m_n^*}\xi} \qquad \textbf{(6.12)}$$

- We then write

$$\boxed{v_{dn} = -\mu_n \xi} \quad \textbf{(6.13)}$$

where

$$\boxed{\mu_n = \frac{q \tau_{cn}}{m_n^*}} \quad \textbf{(6.14)}$$

  - From eqn. (6.13), the electron drift velocity is in opposite direction to the applied electric field.
  - From eqn. (6.14), the electron mobility μ_n is a positive quantity

Table 6.1 list typical mobility values at T = 300 K of some semiconductors with low doping concentration

We see that $\mu_p < \mu_n$, this means for the same electric field, electrons move faster than holes.

Table 6.1

	μ_n (cm²/V-s)	μ_p (cm²/V-s)
Si	1350	480
GaAs	8500	400
Ge	3900	1900

Let us now look at the behavior of the mobility.

- We have earlier seen that charged particles in a semiconductor will be involved in collisions or scatterings with
    - thermally vibrating lattice atoms
    - ionized impurity atoms, and
    - defects

- It is interesting and important to note here, that the thermal vibration of lattice atoms, impurity atoms and defects disrupt the perfect periodic potential in the crystal.
    - The scattering is caused by the imperfection of the periodic potential.
    - A perfect periodic potential will allow the charge carriers (electrons and holes) to move unimpeded, or with no scattering, in the crystal. This can be understood from the wavefunctions discussed in section (3.4.2).
- Let us now discuss the scattering due to lattice vibrations in more detail.
    - Above 0 K, the atoms in a semiconductor vibrate about their lattice position in the crystal.
    - The thermal vibrations cause disruption of the periodic potential, resulting in an interaction between electrons or holes and the vibrating atoms and hence scattering.

- This type of scattering is referred to lattice scattering or phonon scattering.

- The lattice vibrations are expected to increase with increasing temperature, which implies that the number of scatterings per unit time increases. Consequently, the average time between collision decreases and hence, from eqns. (6.9) or (6.14), the mobility decreases.

- Next, let us discuss the scattering due to impurity atoms.

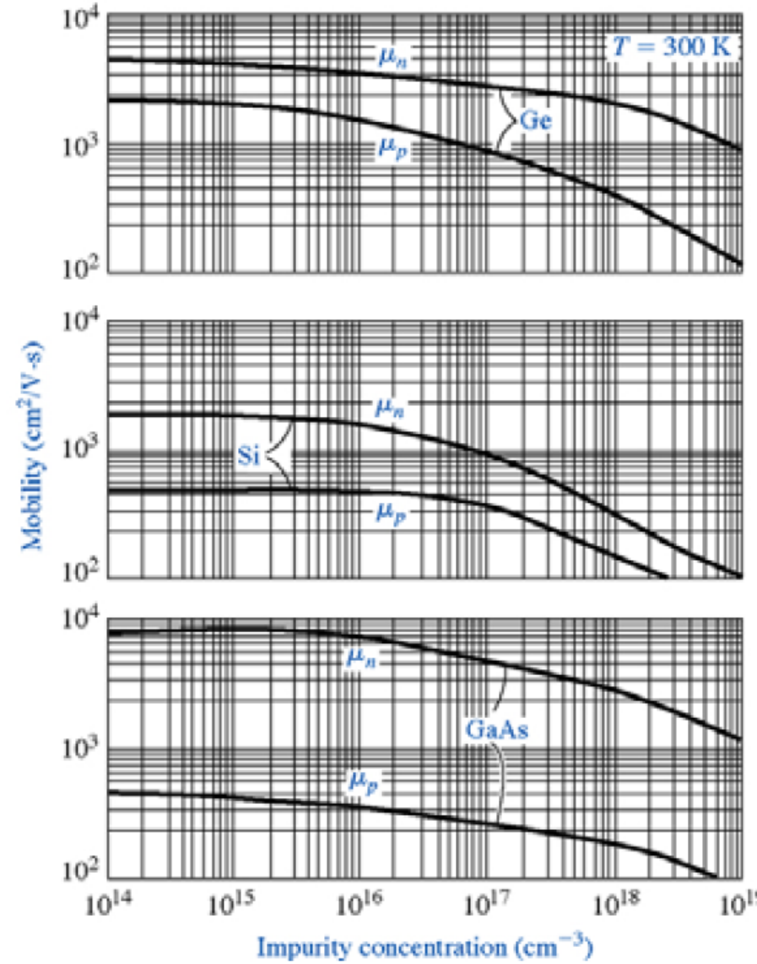
  - As we have seen earlier, the impurity atoms are all ionized at room temperature

  - Again, the ionized impurities disrupt the periodic potential causing the scattering. Here, the interaction between the carriers and the ionized impurities is the coulomb interaction.

  - This scattering is referred to ionized impurity scattering.

- We will expect that the number of scatterings per unit time to increase with increasing impurity concentration, which implies that the average time between scattering decreases. Again, from eqns. (6.9) or (6.14), the mobility will decrease.

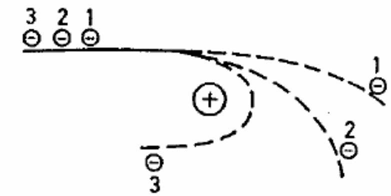
Fig. 6.3 shows the electron and hole mobilities versus impurity concentrations of some semiconductor materials at 300 K.



Electron and hole mobilities versus impurity concentration for Ge, Si and GaAs at 300 K

Fig. 6.3

- We now ask a question, what is the effect of temperature on the impurity scattering.
  - As the temperature increases, the random thermal velocity of carriers increases.
  - This would reduce the time the carriers spend in the vicinity of the ionized impurity center.
  - The less time spent in the vicinity of a coulomb force, the smaller the scattering effect and thus increasing the mobility due to the impurity scattering. Fig. 6.4 shows the schematic of the scattering effect with the carrier velocity.
  - So, in contrast to the lattice scattering, the mobility due to the impurity scattering increases with increasing temperature.



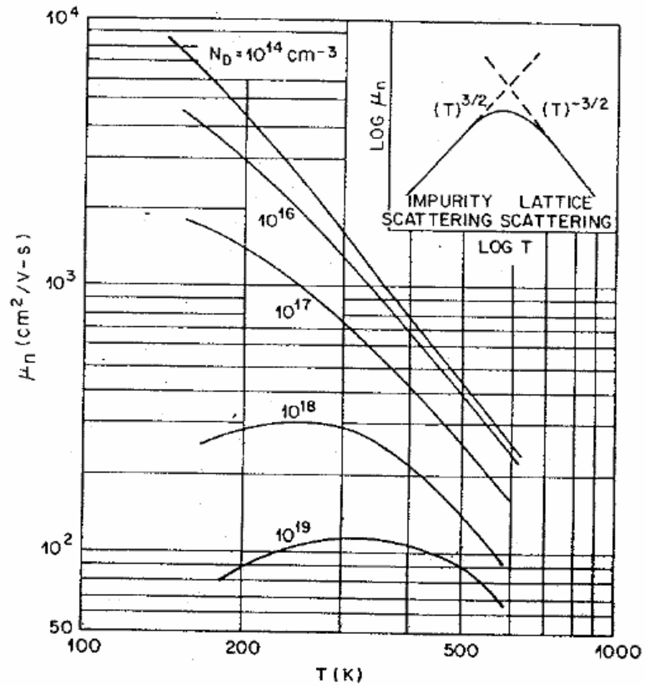
. The carrier scattering by a charged impurity center. Electrons (1, 2 and 3) fly up to the impurity center along the same trajectory but with different velocities. Electron 1 has the greatest velocity; the attraction of the impurity atom in practice does not affect its initial trajectory. Electron 3 has the lowest velocity; the attraction of the charged impurity makes it change its route backwards.

Fig. 6.4

- We have discussed the effect of temperature on the mobility due to the lattice and ionized impurity scatterings.

  Fig. 6.5 shows the electron mobility versus temperature of an n-type semiconductor.

  - We firstly see that the mobility increases as the donor concentration decreases. This behavior has been discussed earlier (see Fig. 6.3).



Electron mobility in silicon versus temperature for various donor concentrations. Insert shows the theoretical temperature dependence of electron mobility.

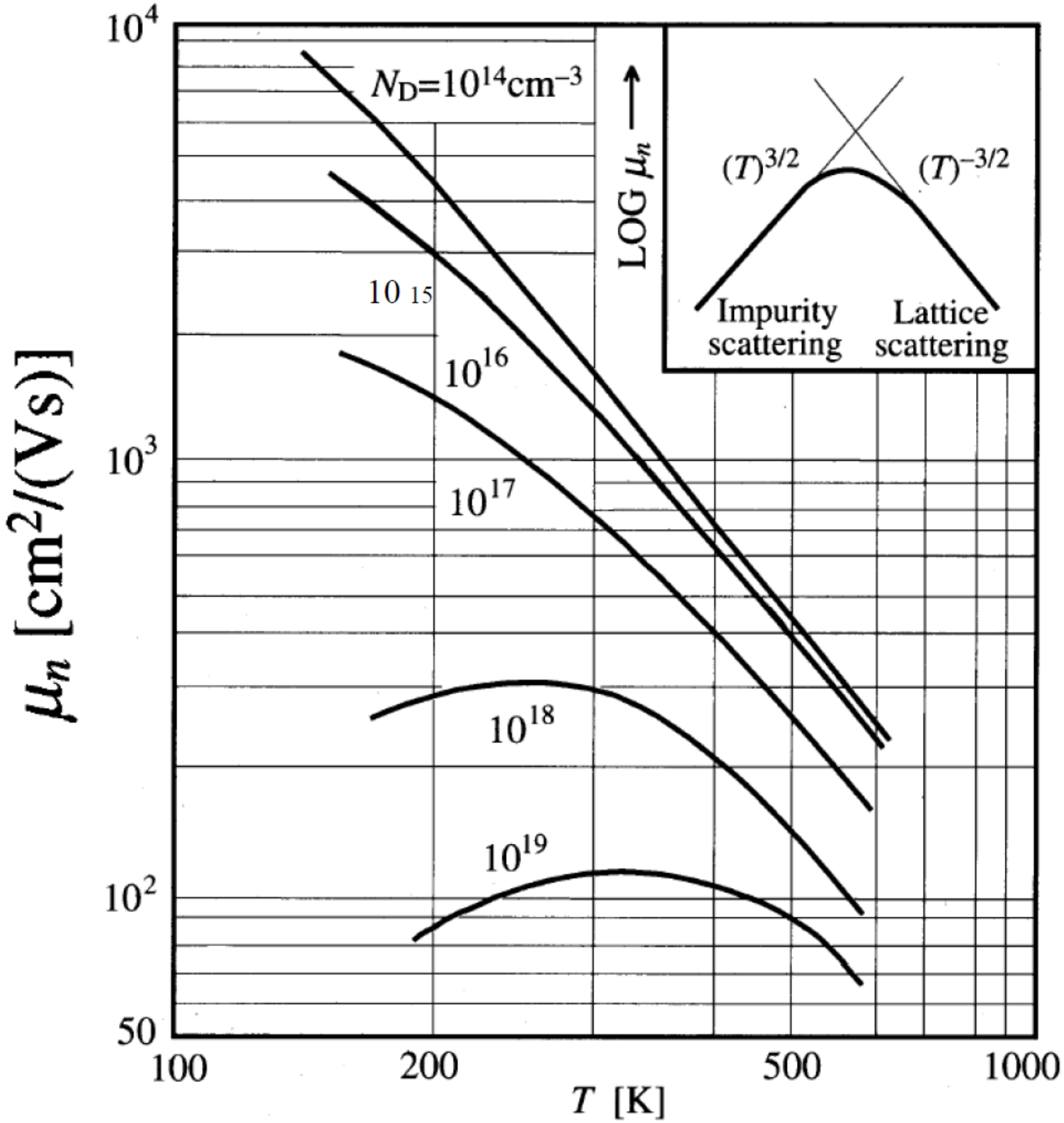
Fig. 6.5

- For each mobility curve, we see that the mobility first increases and then on reaching a maximum begins to fall. So, we can divide into two regions, low and high temperature regions

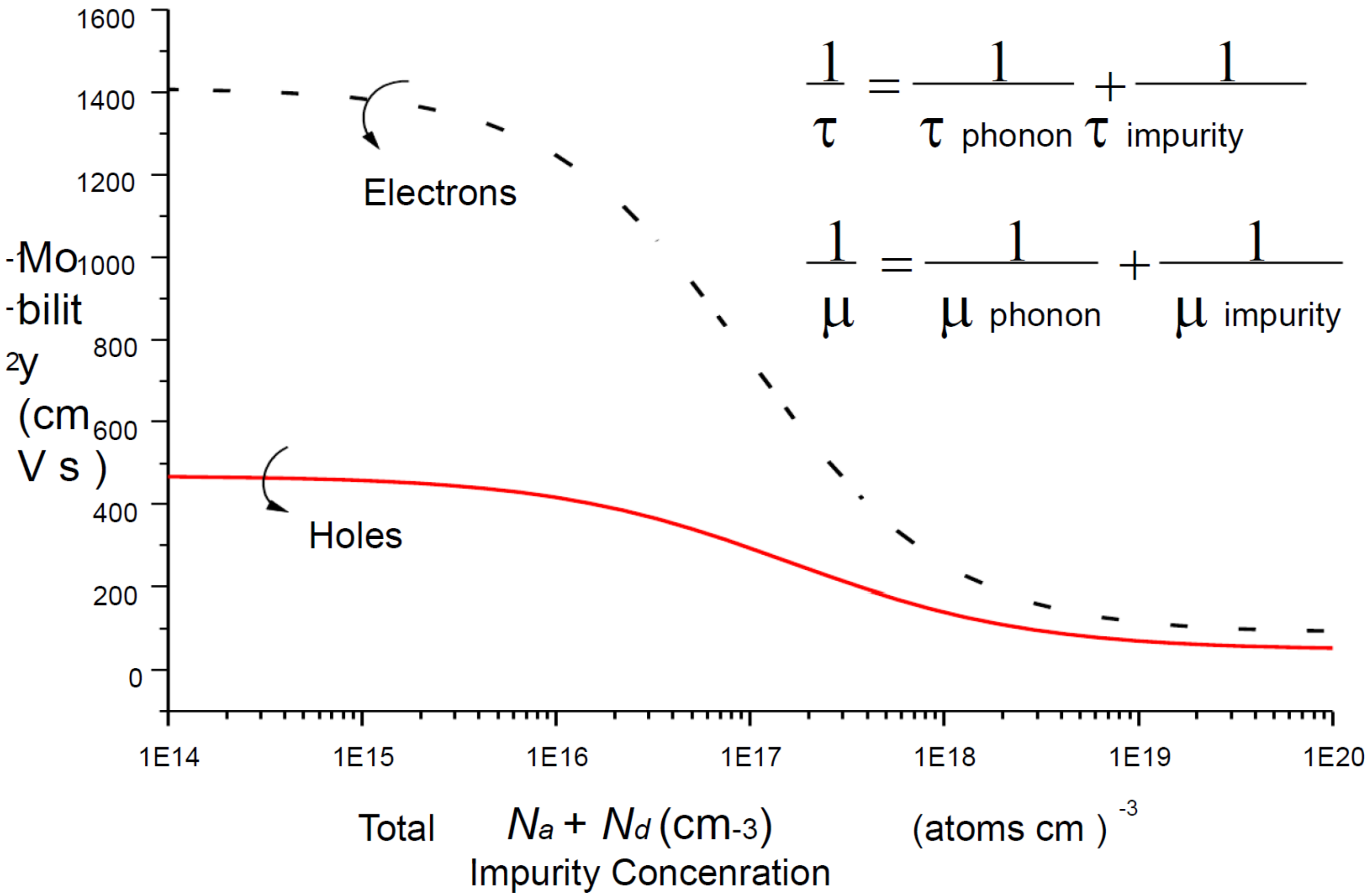
- At high temperature region, the mobility decreases with increasing temperature. This indicates that at this region, the lattice scattering dominates.

- At low temperature region, the mobility increases with increasing temperature, this shows that at this region the impurity scattering dominates.

Temperature Effect on Mobility



Total Mobility



$$\frac{1}{\tau} = \frac{1}{\tau_{\text{phonon}}} + \frac{1}{\tau_{\text{impurity}}}$$

$$\frac{1}{\mu} = \frac{1}{\mu_{\text{phonon}}} + \frac{1}{\mu_{\text{impurity}}}$$

6.2.2 Drift current density

Consider a semiconductor sample with hole density p moving with drift velocity v_{dp} in the presence of an electric field ξ

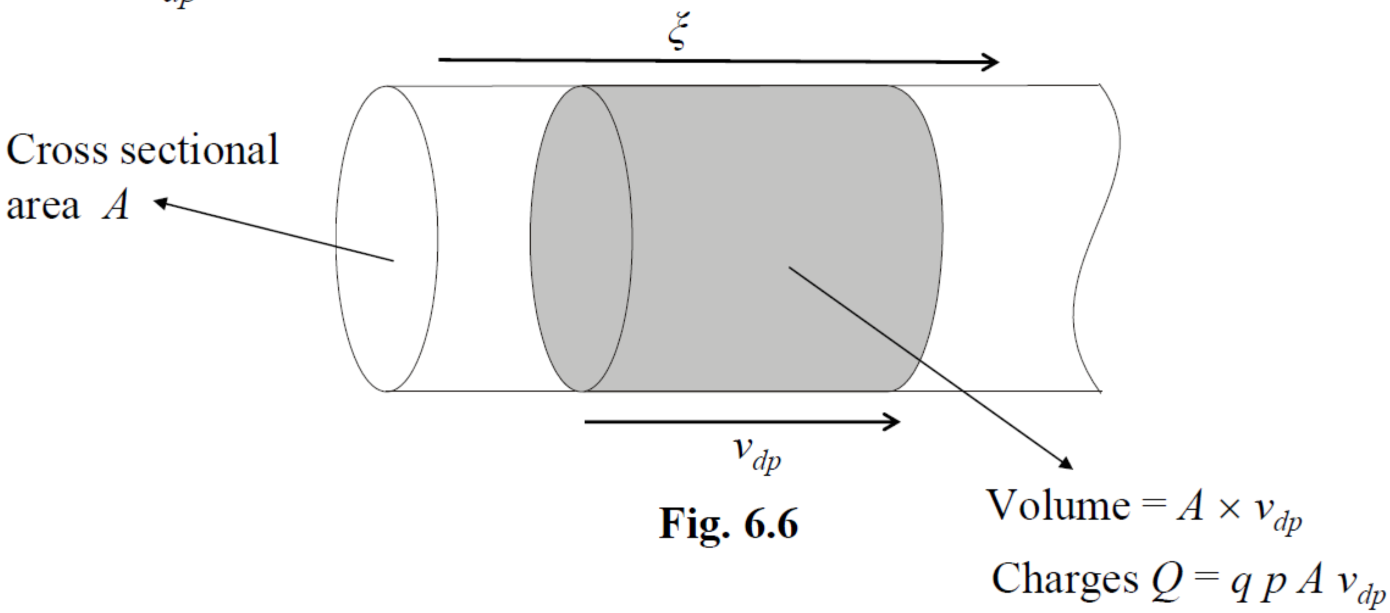


Fig. 6.6

- From Fig. 6.6, the number of charges in the volume $A \times v_{dp}$ is

$$Q = q\ p\ A\, v_{dp} \qquad \textbf{(6.15)}$$

- Since the drift velocity is v_{dp}, in one second, the charges Q in the volume will cross the cross sectional A.
  - By the definition of a current : the number of charge crossing an area per second
  - It follows that the hole drift current

$$I_{p\ drift} = q\ p\ A\ v_{dp} \qquad \textbf{(6.16)}$$

  - Hence, the hole drift current density is

$$J_{p\ drift} = q\ p\ v_{dp} \qquad \textbf{(6.17)}$$

Remember, the current density is the number of charge crossing a unit area per second.

- Using eqn. (6.8), we can substitute v_{dp} in eqn. (6.17). We then obtain

$$J_{p\ drift} = q\ p\ \mu_p\ \xi \qquad \textbf{(6.18)}$$

We can apply the same analysis for electrons.

- If we do that we obtain the electron drift current density

$$J_{n\ drift} = -\ q\ n\ v_{dn} \qquad \textbf{(6.19.a)}$$

Here, the negative (– ve) sign is because the electron is negatively charged. The electron drift current density $J_{n\ drift}$ is in the opposite direction to the electron drift velocity v_{dn}.

- From the expression for v_{dn} in eqn. (6.13), we get

$$J_{n\,drift} = -q\,n\left(-\mu_n\,\xi\right) = q\,n\,\mu_n\,\xi \qquad \textbf{(6.19.b)}$$

- In semiconductors the electrical conduction is due to both electron and hole currents.
  - Therefore, the total drift current density J_{drift} is the sum of the individual electron and hole drift current densities

$$J_{drift} = J_{n\,drift} + J_{p\,drift}$$

$$J_{drift} = q\,\mu_n\,n\,\xi + q\,\mu_p\,p\,\xi = \left(q\,\mu_n\,n + q\,\mu_p\,p\right)\xi \qquad \textbf{(6.20)}$$

- We may write the total drift current density J_{drift} in eqn. (6.20) as

$$\boxed{J_{drift} = \sigma\,\xi} \qquad \textbf{(6.21)}$$

where

$$\boxed{\sigma = q\mu_n n + q\mu_p p} \qquad \textbf{(6.22)}$$

  - σ is the electrical conductivity of the semiconductor material.
  - the unit is (ohm-cm)$^{-1}$ or (Ω-cm)$^{-1}$

- Resistivity ρ is the reciprocal of conductivity

$$\boxed{\rho = \frac{1}{\sigma} = \frac{1}{q\mu_n n + q\mu_p p}} \qquad \textbf{(6.23)}$$

  - the unit is (ohm-cm) or (Ω-cm)

- For the case of an extrinsic semiconductor, where the majority carrier concentration greatly exceeds the minority carrier concentration, we can write as follow
  - For p-type where $p >> n$, $$\rho = \frac{1}{\sigma} = \frac{1}{q\mu_p p} \quad \textbf{(6.24.a)}$$
  - For n-type where $n >> p$, $$\rho = \frac{1}{\sigma} = \frac{1}{q\mu_n n} \quad \textbf{(6.24.b)}$$

Example 6.1

Consider a bar of semiconductor as shown in Fig. 6.7 with an applied voltage V that produces a current I. Using eqn. (6.21) derive the expression $V = IR$ where R is the resistance of the semiconductor.

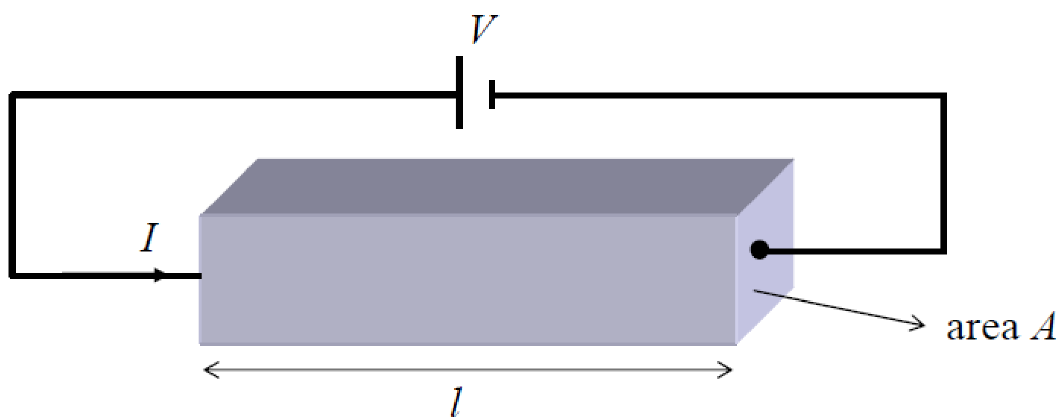


Fig. 6.7

The current density is $J = \dfrac{I}{A}$ and the electric field is $\xi = \dfrac{V}{l}$

From eqn. (6.21) $J = \sigma \xi$

$$\frac{I}{A} = \sigma \frac{V}{l}$$

$$V = \left(\frac{1}{\sigma} \frac{l}{A} \right) I$$

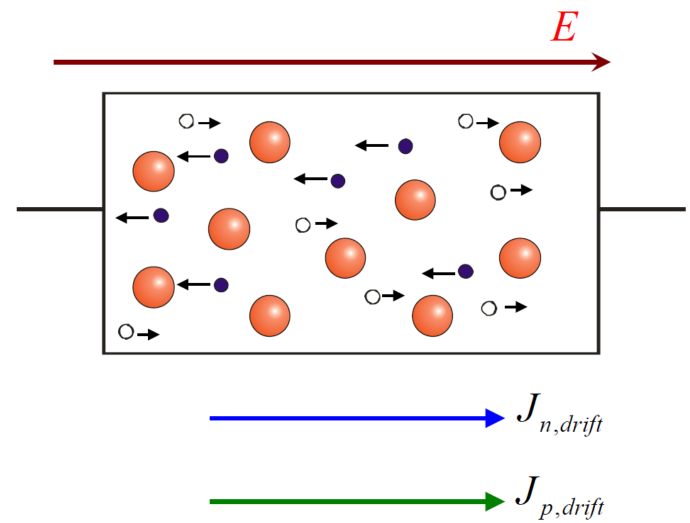
$$V = \left(\rho \frac{l}{A} \right) I = IR$$ which is Ohm's law

where $R = \rho \dfrac{l}{A}$

Drift Current and Conductivity

$$J_{p,drift} = qpv = qp\mu_p E$$

$$J_{n,drift} = -qnv = qn\mu_n E$$



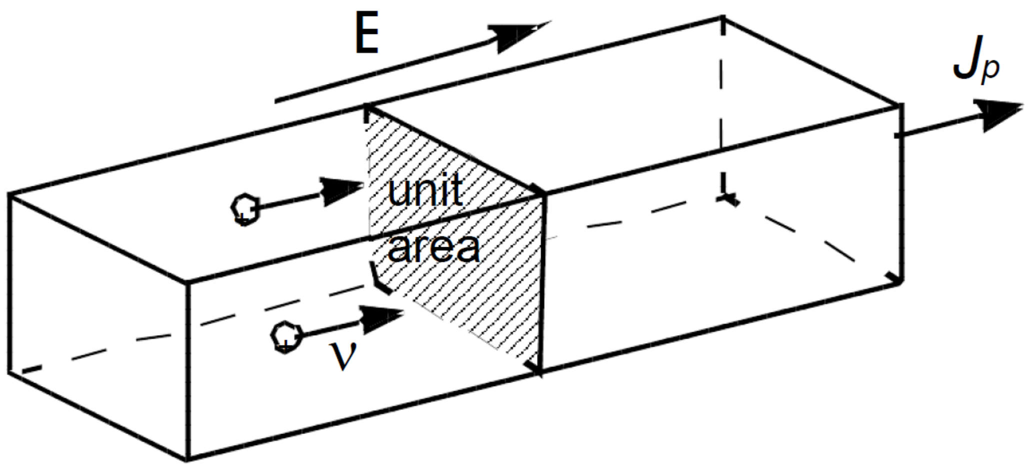
$$J_{drift} = J_{n,drift} + J_{p,drift} = \sigma\, E = (qn\mu_n + qp\mu_p)E$$

$\therefore$ ***conductivity*** *(1/ohm-cm) of a semiconductor is*

$$\sigma = qn\mu_n + qp\mu_p$$

$1/\sigma$ = is resistivity (ohm-cm)

Drift Current and Conductivity



Hole current density $\boxed{J_p = qpv}$ A/cm² or C/cm²·sec

EXAMPLE: If $p = 10^{15}\text{cm}^{-3}$ and $v = 10^4$ cm/s, then

$$J_p = 1.6\times10^{-19}\text{C} \times 10^{15}\text{cm}^{-3} \times 10^4\text{cm/s}$$

$$= 1.6\ \text{C/s}\cdot\text{cm}^2 = 1.6\ \text{A/cm}^2$$

Electron mobility example

The electron mobility in silicon (Si) is 1000 $cm^2/(V\text{-}s)$.

What is the electron drift velocity if the voltage across the sample is 5V and the sample is d =1 mm thick?

Solution. Assuming **uniform** electric field in the sample, the electric field and the voltage are related as: $\boldsymbol{E = V/d}$, where d is the distance between the electrodes.

$E = V/d = 5V/10^{-3}\ m = 5 \cdot 10^3\ V/m$

$v_{dr} = \mu\ E$; $\mu = 1000\ cm^2/(V\text{-}s) = 1000 \times 10^{-4}\ m^2/(V\text{-}s) = 0.1\ m^2/(V\text{-}s)$.

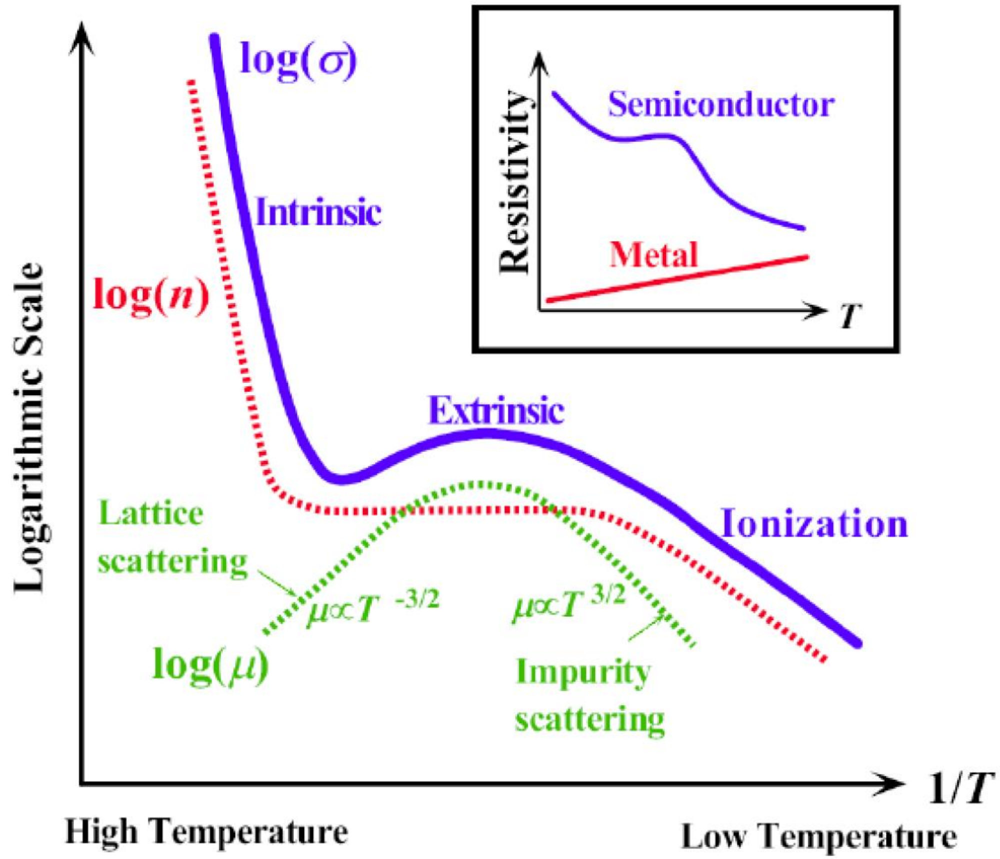
$v_{dr} = 0.1\ m^2/(V\text{-}s) \times 5 \cdot 10^3 V/m = 500\ m/s$

Compare this to the electron "thermal" velocity: $v_T \approx 10^5\ m/s >> v_{dr}$

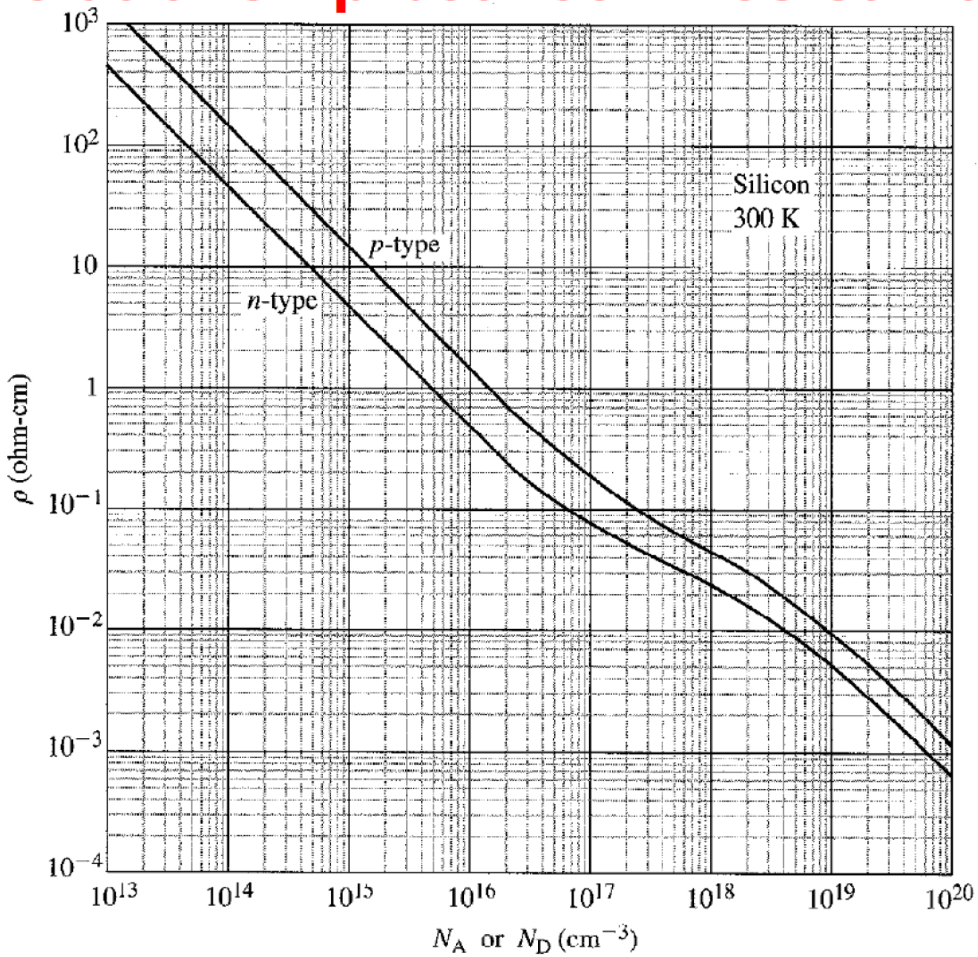
Conductivity temperature dependence

Conductivity dependen on both carrier conceration and mobility, both are a function of T

$$\sigma = en_i(u_e + u_h)$$



Relationship between Resistivity and Dopant Density



conductivity

$\sigma = qn\mu_n + qp\mu_p$

resistivity

$\rho = 1/\sigma$

- Note that resistivity changes by 7 orders of magnitude with doping
- Slope changes from linear to sub-linear ?

Quiz :

1). Doping always increases the conductivity?

2). Intrinsic semiconductors have the lowest conductivity?

Ohm's "Law"

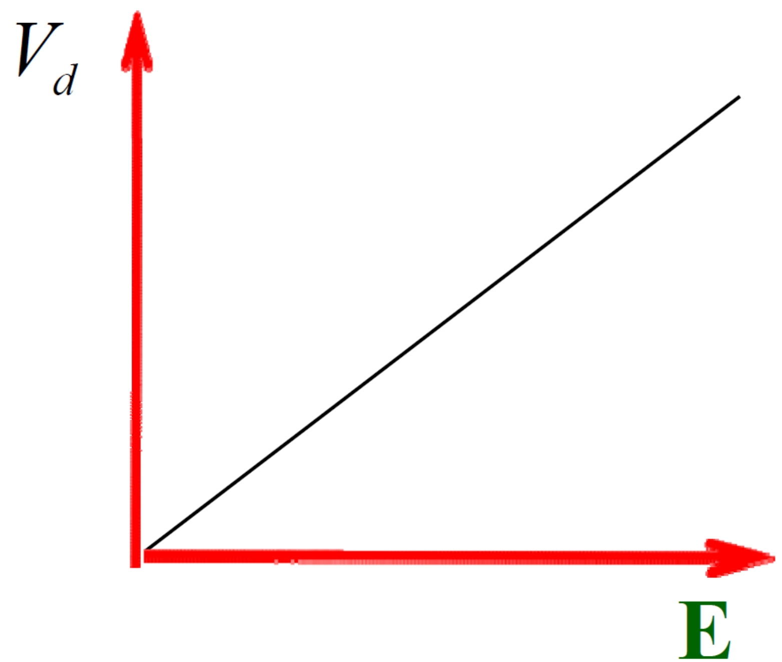
- This says the **Drift Velocity V_d** is **linear in the electric field E:** $V_d = \mu E$

$\mu \equiv$ Mobility

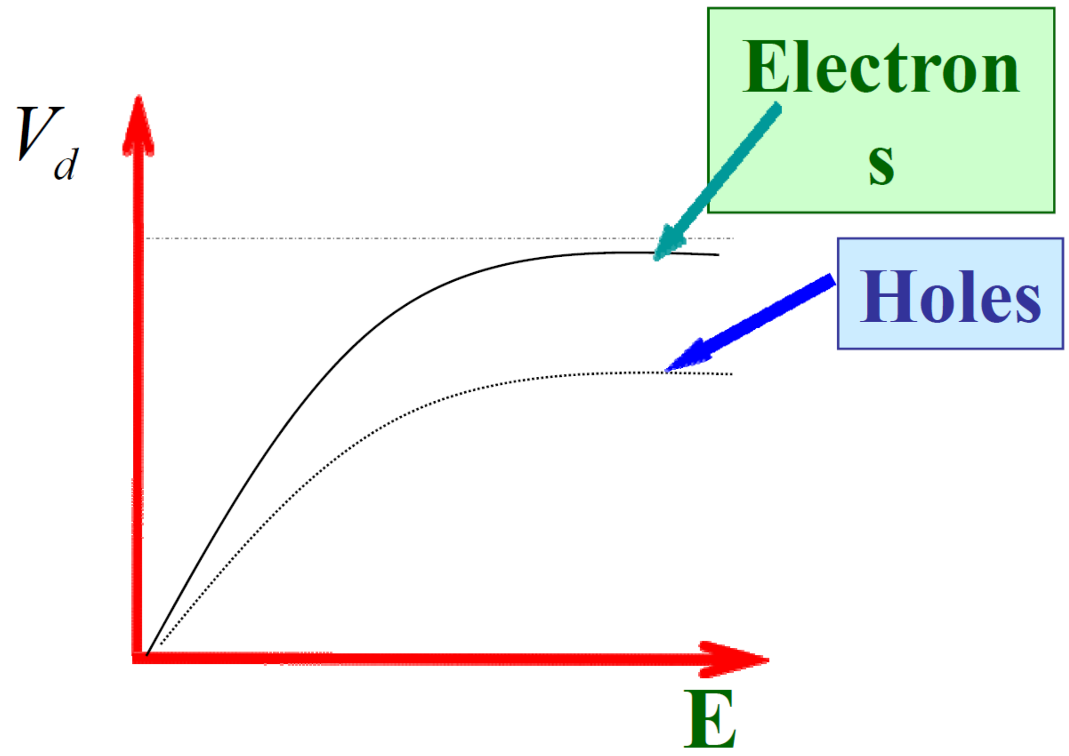
- If this were true for all **E**, the charge carriers could be made to go fast without limit, just by increasing **E**! That would be nonsense! So, in every material, ***at high enough*** **E**, the **V_d vs E curve** ***must saturate to a constant value!***

Ohm's "Law" $V_d = \mu E$

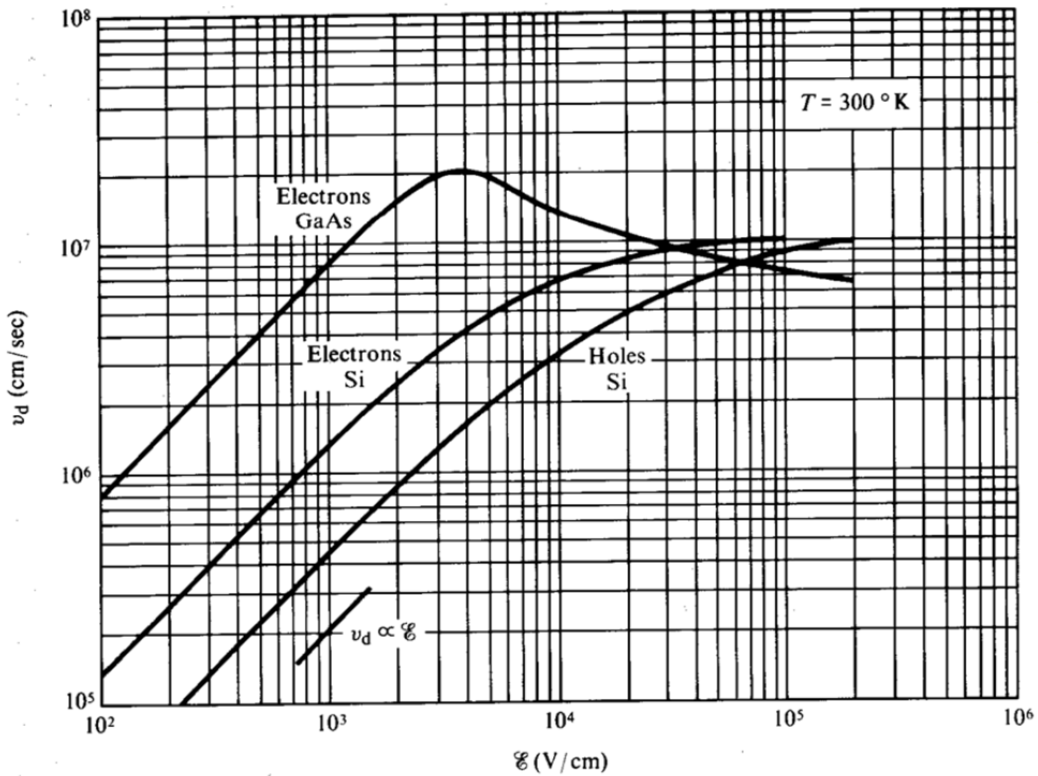
- Obviously, this says that the **V_d vs E curve** looks qualitatively like:



- Measurement shows that, in all materials, at high enough **E**, the **V_d vs E curve** looks qualitatively like:



E Field Dependence of Drift the Velocity



Velocity Saturation

In n-type Si, the saturation velocity

$$V_s \sim 10^7 \text{cm/s}$$

at a field

$$E_s \sim 10^4 \text{ V/cm}$$

The **carrier velocity saturation** at high **E** fields clearly ***places a FUNDAMENTAL upper limit on the speed*** of semiconductor devices.

Why velocity is saturated at large E?

For a certain semiconductor, mean free path L is a constant, for example, for Si, L~=1 um, and thus mean free time：

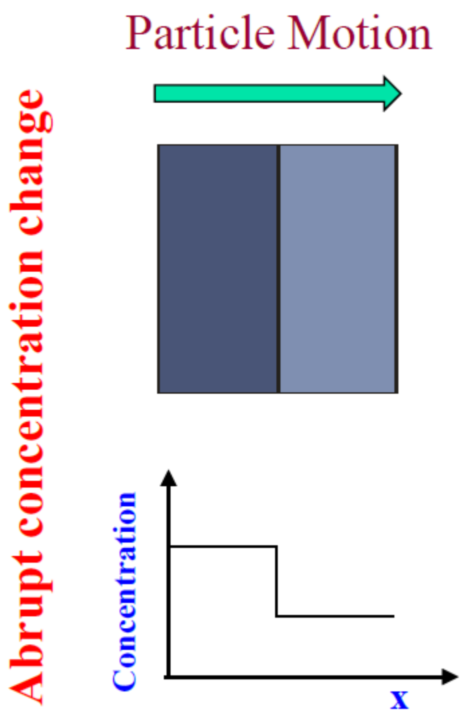
$\tau_0 = 1\mu m/Vth = 1psec$, where $Vth = 10^7$ cm/sec is the thermal motion velocity.

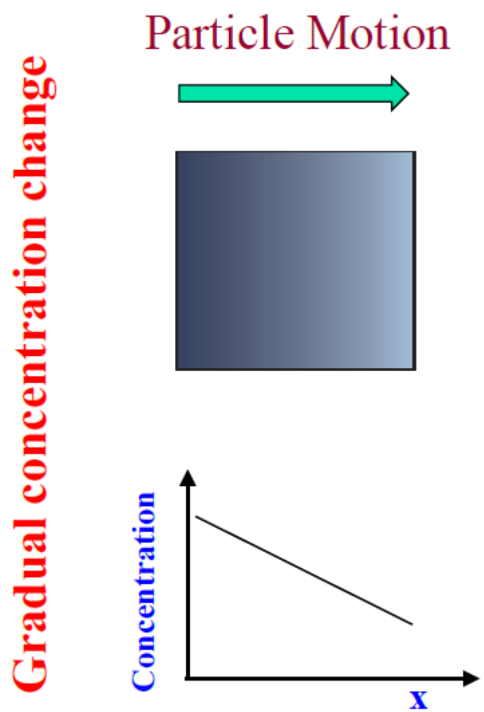
As E is increased, $V = \mu_0 E > Vth$, then $\tau = L/V < \tau_0$ and thus $\mu < \mu_0$, *which in turn tend to reduce the increasing of V (because V=uE, u is reduced when E is increased to a certain limit).*

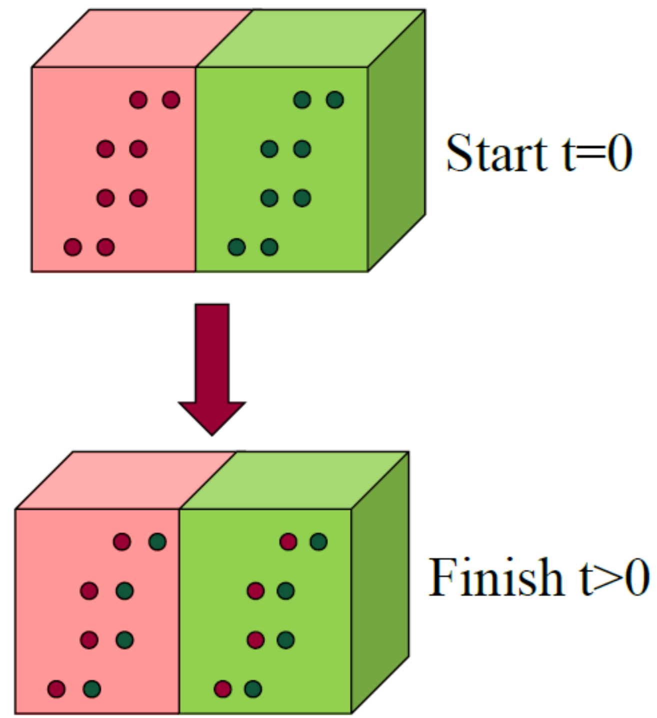
Diffusion Current

Diffusion is due to concentration difference between two regions of a semiconductor

The carriers will move from higher concentration region to the lower one.







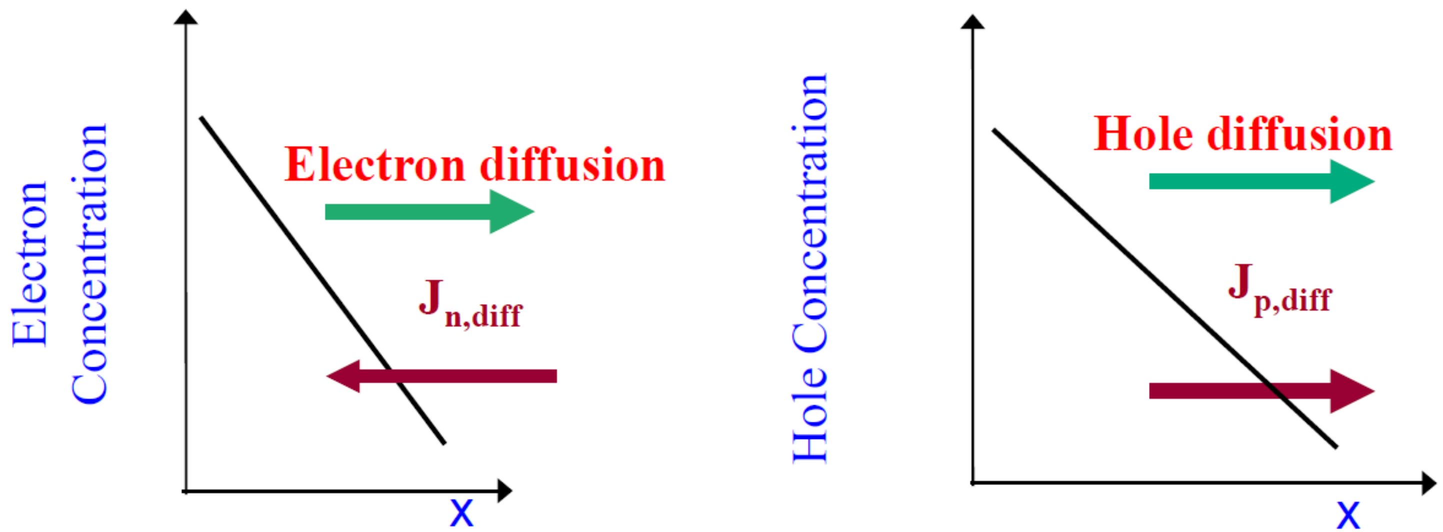
…continued… Diffusion Current

The electron diffusion current density: $$J_{n,diff} = qD_n \frac{dn}{dx}$$

D_n is the diffusion coefficient of electrons

The hole diffusion current density: $$J_{p,diff} = qD_p \frac{dp}{dx}$$

D_p is the diffusion coefficient of holes



Total Current – Review of 4 Current Components

$$\boxed{J_{TOTAL} = J_n + J_p}$$

$$J_n = J_{n,drift} + J_{n,diffusion} = qn\mu_n \mathbf{E} + qD_n \frac{dn}{dx}$$

$$J_p = J_{p,drift} + J_{p,diffusion} = qp\mu_p \mathbf{E} - qD_p \frac{dp}{dx}$$

The Einstein Relation

Einstein showed that diffusion coefficient of any particle and its mobility are not independent parameters. They are related by

$$\boxed{D_p = \frac{k_B T}{q} \mu_p}$$

$$\boxed{D_n = \frac{k_B T}{q} \mu_n}$$

Table 6.2 lists typical mobility and diffusion coefficient values of electrons and holes at T = 300 K (mobility in cm^2/V-sec and diffusion coefficient in cm^2/sec)

Table 6.2

	μ_n	D_n	μ_p	D_p
Silicon	1350	35	480	12.4
Gallium Arsenide	8500	220	400	10.4
Germanium	3900	101	1900	49.2

Summary Equations

$$v_p = \mu_p \mathbf{E}$$

$$v_n = -\mu_n \mathbf{E}$$

$$J_{p,drift} = qp\mu_p \mathbf{E}$$

$$J_{n,drift} = qn\mu_n \mathbf{E}$$

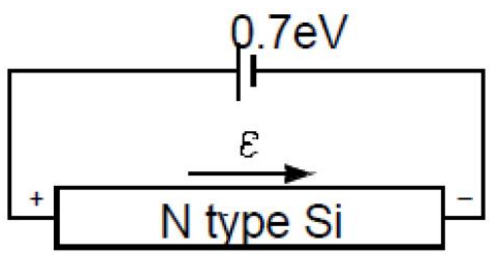
$$J_{n,diffusion} = qD_n \frac{dn}{dx}$$

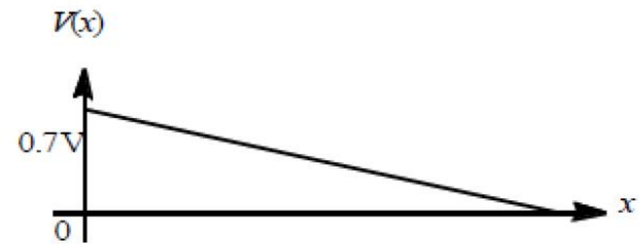
$$J_{p,diffusion} = -qD_p \frac{dp}{dx}$$

$$D_n = \frac{kT}{q}\mu_n$$

$$D_p = \frac{kT}{q}\mu_p$$

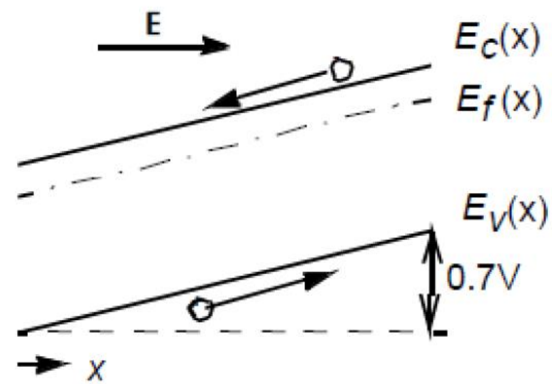
Relation Between the Energy Diagram and V, E





E_c and E_v vary in the opposite direction from the voltage. That is, E_c and E_v are higher where the voltage is lower.

$$\mathbf{E}(x) = -\frac{dV}{dx} = \frac{1}{q}\frac{dE_c}{dx} = \frac{1}{q}\frac{dE_v}{dx}$$



Electrical field E=-dV/dx
Electrical energy E=-qV

能带的高度=>电压
能带的斜率=>电场
电场的斜率=>电荷

In 1-D, the electric field is related to potential by the relation

$$\mathcal{E} = -\frac{dV}{dx}$$

Therefore, $\mathcal{E} = \frac{1}{q}\frac{dE_c}{dx} = \frac{1}{q}\frac{dE_i}{dx} = \frac{1}{q}\frac{dE_v}{dx}$

Semiconductor in Thermal Equilibrium

When a semiconductor is in equilibrium, there's no *net* flux

Since $F_n = D_n \frac{dn}{dx} + \mu_n n \mathcal{E} = 0$

$$n = n_i e^{(E_F - E_i)/kT}$$

$$\frac{dn}{dx} = \frac{n}{kT}\left[\frac{dE_F}{dx} - \frac{dE_i}{dx}\right]$$

Then

$$F_n = \frac{D_n}{kT} n \left[\frac{dE_F}{dx} - \frac{dE_i}{dx} \right] + \mu_n n \frac{1}{q} \frac{dE_i}{dx}$$

Using the Einstein relationship,

$$\frac{D_n}{kT} = \frac{\mu_n}{q}$$

$$F_n = \frac{D_n}{kT} n \frac{dE_F}{dx}$$

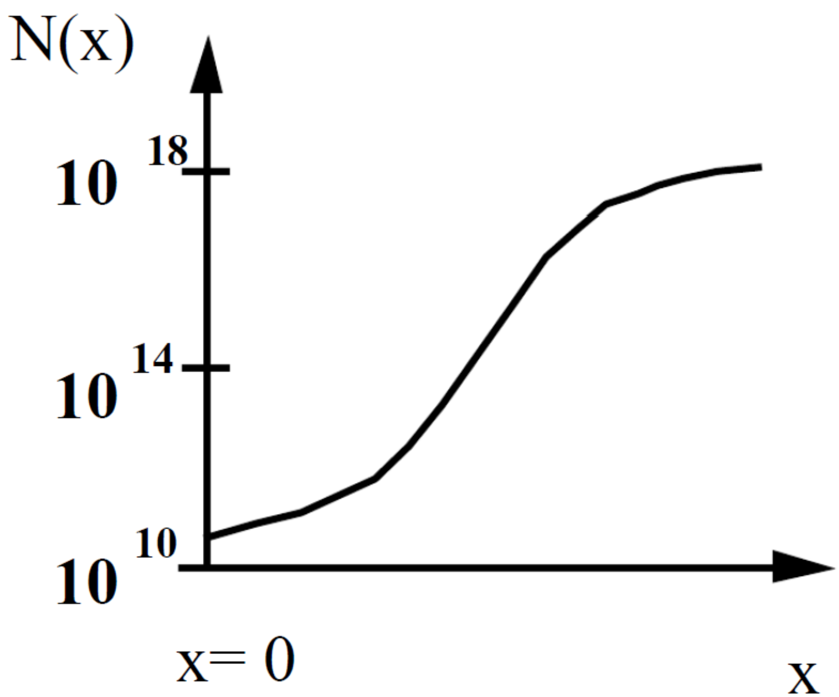
A gradient in the Fermi level is the driving force for carrier motion

In equilibrium, $F_n = 0 \Rightarrow$ $\boldsymbol{E_F}$ **= constant.**

This is an extremely important result!

Non-Uniform Doping Distributions

We have assumed in the previous discussion that the doping levels are spatially uniform. Often, this is not the case. Consider the simple situation where we have a n-type semiconductor with a non-uniform $N_D(x)$.



The impurity profile drawn represents the fixed position of ionized dopants. The carriers are mobile and will attempt to diffuse away from regions of high concentrations. As the electrons move away (diffuse) from the high concentration region, they *leave behind positively charged donors* setting up an electric field to pull the electrons back (drift).

Equilibrium is established when the diffusion and drift cancel. Hence $n(x)$ is not *exactly* the same as $N_D(x)$ and a "built-in" electric field exists.

In equilibrium,

$$F_n(x) = D_n \frac{dn(x)}{dx} + \mu_n n(x)\mathcal{E}(x) = 0$$

$$\mathcal{E}(x) = -\frac{D_n}{\mu_n n(x)} \frac{d\,n(x)}{dx}$$

$$= -\frac{kT}{q} \frac{1}{n(x)} \frac{dn(x)}{dx} \qquad \text{(Using Einstein relationship)}$$

Use Possion's equation,

$$\frac{d\mathcal{E}(x)}{dx} = \frac{\rho(x)}{K_s \varepsilon_o}$$

Where

$\rho(x)$ = q [$N_D(x)$ - $n(x)$] = space charge density (coul cm^{-3})

K_s = dielectric constant

ε_o = permittivity of free-space = 8.85 ´ 10^{-14} F/cm

$n(x)$ and $\mathcal{E}(x)$ can be solved.

If the concentration gradient is very gradual, the built in electric field is very small, and ρ is very small, compared with N_D. We can assume that the semiconductor is locally $\approx$ neutral, which means $\rho(x) \approx 0$, $n(x) \approx N_D(x)$.

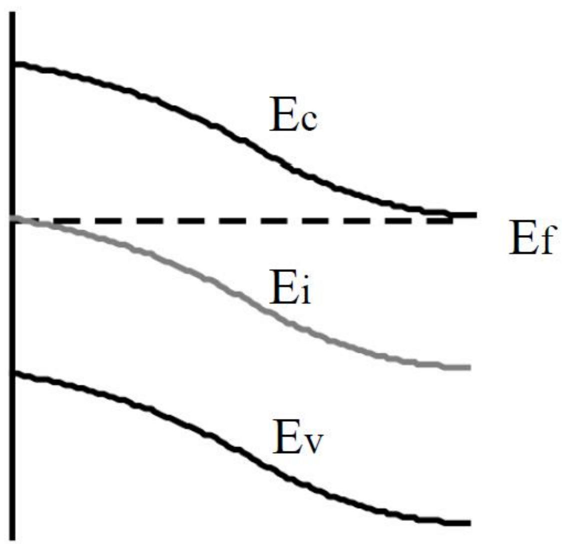
This is the "QUASI-NEUTRALITY" assumption.

To construct the band diagram for an inhomogeneously doped semiconductor, we start with $\boldsymbol{E_F}$ **= constant.**

We now assume "QUASI-NEUTRALITY".

That is, $n(x) \approx N_D(x)$.

$$E_F - E_i(x) \approx \ kT \ln\left[\frac{N_D(x)}{n_i}\right] = 60meV \log_{10}\left[\frac{N_D(x)}{n_i}\right]$$



Internal electric field induced band bending

$$\mathbf{E}(x) = -\frac{dV}{dx} = \frac{1}{q}\frac{dE_c}{dx} = \frac{1}{q}\frac{dE_v}{dx}$$

Quiz

For below band diagrams, what are the differences between them?
Specify the differences in terms of doping, electrical field, equilibrium etc.